Knowledge Graph Construction, Alignment, Reasoning & RAG

Web Mining & Semantics — Final Project Report

Domain: Football / FIFA Governance & Institutional Documentation

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1. Data Acquisition & Information Extraction

1.1 Domain & Seed URLs

The chosen domain is Football governance and FIFA institutional documentation. The pipeline targets official FIFA publications, laws of the game, and encyclopedic sources covering FIFA's structure, regulations, and stadium guidelines.

8 seed URLs were selected and validated:

- https://publications.fifa.com/fr/football-stadiums-guidelines/introduction/
- https://publications.fifa.com/fr/.../stadium-guidelines/information-technology/
- https://en.wikipedia.org/wiki/FIFA
- https://en.wikipedia.org/wiki/Laws_of_the_Game_(association_football)
- https://en.wikipedia.org/wiki/FIFA_World_Cup
- https://en.wikipedia.org/wiki/International_Football_Association_Board
- https://en.wikipedia.org/wiki/Video_assistant_referee
- https://en.wikipedia.org/wiki/UEFA

1.2 Crawler Design & Ethics

The crawler uses trafilatura to extract the main content zone from each URL, removing navigation menus, footers, and boilerplate. robots.txt compliance was implemented via Python's built-in RobotFileParser. Wikipedia returns HTTP 403 on its robots.txt endpoint — Python interprets this as disallow-all per RFC 9309. Since Wikipedia's Terms of Use explicitly permit non-commercial academic crawling, these domains were whitelisted manually with documented justification.

Domain	robots.txt Status	Resolution
publications.fifa.com	Allow: /	Crawled normally
en.wikipedia.org	HTTP 403 on robots.txt	Whitelisted — Wikipedia ToU permits academic use

1.3 Cleaning Pipeline & NER Results

Text extraction uses trafilatura's extract() function. NER is performed with spaCy (en_core_web_sm). Target entity types: PERSON, ORG, GPE. DATE entities are excluded from the URI space and treated as literals.

Metric	Value
Pages crawled	8
Total entities extracted	2,188
ORG	912
PERSON	624
GPE	416
DATE (excluded)	236

1.4 Entity Ambiguity — 3 Cases

Entity	Extracted	Correct	Explanation
matchday	DATE	Concept	Temporal suffix triggers DATE classifier. It is a football-specific operational concept, not a calendar date.

Entity	Extracted	Correct	Explanation
Olympique Lyonnais	PERSON	ORG	Appears in credits list without context. Two-word proper noun pattern matches PERSON classifier.
Jaaf Design	PERSON	ORG	Graphic design company. Absence of 'Ltd/Inc' markers leads to PERSON misclassification.

2. KB Construction, Alignment & Expansion

2.1 RDF Modeling Choices

The private KB uses namespace http://example.org/fifa/ (prefix ex:). Entities are typed as ex:Person, ex:Organization, or ex:Place with rdfs:label literals. Relational triples use lemmatized verbs as predicate URIs. Three ontological classes link to schema.org equivalents via rdfs:subClassOf.

```
ex:Organization  rdfs:subClassOf  schema:Organization .
ex:Person        rdfs:subClassOf  schema:Person .
ex:Place         rdfs:subClassOf  schema:Place .
```

2.2 Entity Linking

Entity linking uses the Wikidata wbsearchentities API on the top 50 most frequent entities. Confidence: 0.9 for exact label match, 0.6 for partial. Threshold: 0.7. 35 entities linked; 15 declared as new private entities.

Private Entity	Wikidata URI	Confidence	Status
ex:fifa	wd:Q10400	0.9	Linked
ex:uefa	wd:Q349	0.9	Linked
ex:gianni_infantino	wd:Q313911	0.9	Linked
ex:world_cup	wd:Q19317	0.9	Linked
ex:switzerland	wd:Q39	0.9	Linked
ex:london	wd:Q84	0.9	Linked

2.3 Predicate Alignment

Private Predicate	Wikidata Property	Label
ex:win	wdt:P166	award received
ex:organise	wdt:P664	organizer of
ex:include	wdt:P527	has part
ex:hold	wdt:P664	organizer of
ex:represent	wdt:P1268	represents
ex:found	wdt:P571	inception
ex:govern	wdt:P748	appointed by
ex:comprise	wdt:P527	has part

2.4 Final KB Statistics

Metric	Value	Target (TD4)
Total triples	19,515	50,000 – 200,000
Entities	8,936	5,000 – 30,000 ✓
Relations	1,729	50 – 200
Linked entities	35	—
Aligned predicates	8	—

Note : *Triple count below target due to limited expansion anchors (35 entities). Entity count is within range. Relation count exceeds target — discussed in Section 6.*

3. Knowledge Reasoning with SWRL

3.1 SWRL Rule on family.owl

The following SWRL rule infers that any Person older than 60 is an oldPerson:

```
Person(?p) ^ hasAge(?p, ?a) ^ swrlb:greaterThan(?a, 60) -> oldPerson(?p)
```

owlready2 does not support SWRL numeric built-ins via set_as_rule(). The rule was applied programmatically in Python with equivalent semantics.

Individual	Age	Inferred as oldPerson?
Alice	72	Yes
Carol	61	Yes
Bob	35	No

3.2 SWRL Rule on the FIFA KB

```
Organization(?o) ^ hasOrganized(?o, ?e) -> isFIFAAffiliated(?o)
```

Applied manually on the expanded KB via the ex:organise predicate. This identifies organizations that organized FIFA-related events as FIFA-affiliated.

3.3 Rule-Based vs Embedding-Based Reasoning (Exercise 8)

The equivalent TransE vector arithmetic for the rule above would be: vector(org) + vector(ex:organise) ≈ vector(affiliated_org). On a 19k-triple KB, the embedding-based result is less reliable than the rule due to insufficient training density. Rules are exact but require manual design; embeddings generalize but need dense training data.

4. Knowledge Graph Embeddings

4.1 Data Splits

Split	Triples	Percentage
Train	~15,612	80%
Validation	~1,951	10%
Test	~1,952	10%

4.2 Training Configuration

Hyperparameter	Value
Embedding dimension	100
Learning rate	0.01 (Adam)
Batch size	512
Epochs	100
Negative sampler	basic
Framework	PyKEEN

4.3 Evaluation Results

Model	MRR	Hits@1	Hits@3	Hits@10
TransE	0.1395	0.0408	0.1674	0.3498
DistMult	0.1972	0.1137	0.2232	0.3712

DistMult outperforms TransE on all metrics. The FIFA domain contains predominantly symmetric relations (membership, participation), which align with DistMult's symmetric scoring function. TransE's translational model is penalized by symmetric patterns.

4.4 KB Size Sensitivity

Subset	Triples	Observation
25% (~4k)	~3,903	Unstable — insufficient entity coverage
50% (~8k)	~7,806	Partial structure learning
100% (~16k)	~15,612	Best available — MRR 0.1395, Hits@10 0.3498

4.5 Embedding Analysis

t-SNE clustering on 500 sampled entities shows partial separation by ontological class. Place entities cluster more distinctly; Organization and Person entities overlap — consistent with heterogeneous Wikidata expansion triples blurring class boundaries.

Note : *Insert tsne_clustering.png here — generated by TD5 notebook Cell 14.*

5. RAG over RDF/SPARQL

5.1 Setup

Machine: MacBook Air. Model: llama3.2:1b via Ollama (http://localhost:11434). Graph: expanded_kb.nt (19,515 triples). Framework: rdflib 6.x for SPARQL execution.

5.2 Schema Summary & Prompt Design

The schema summary is built automatically: namespace prefixes, 10 private predicates (ex: namespace), 6 classes, and 5 sample triples from the private namespace. The summary is intentionally minimal to prevent the LLM from copying schema content into generated queries — a problem observed with larger schema summaries. The self-repair loop submits failed queries back with the error message for one correction attempt.

5.3 Evaluation Results

Q	Question	Baseline	Gold SPARQL	LLM RAG	Correct
Q1	Which organizations are in the KB?	Generic — unaware of KB	15 rows (ESPN Inc., FIFA WG...)	ERROR: 'EXAFI' at char 0	No
Q2	Which persons are mentioned?	Generic FIFA database answer	15 rows (Andrew, J. Lillywhite...)	ERROR: 'ex' at char 25	No
Q3	Which places are referenced?	Cannot specify without context	15 rows (N. Berwick, Baku...)	ERROR: 'EXTRACT' at char 0	No
Q4	What is FIFA linked to?	Correct general answer from memory	15 rows (indict → US DOJ)	ERROR: 'EX' at char 0	No
Q5	Entities linked to London?	Generic London answer	11 rows (fifa, exclude...)	ERROR: 'SKOS' at char 0	No

Note : *Gold SPARQL = manually validated reference queries. These confirm the KB contains the requested data. All LLM failures are generation failures, not KB failures.*

5.4 Screenshots

The following screenshots show the complete evaluation run (eval command) on the FIFA RAG chatbot:

```
~/Doc/Webdata mining /Lab 2   INT ✗   32s   base   16:08:30
└─ python lab_rag_sparql_gen.py
[INFO] Loaded 19515 triples from expanded_kb.nt

============================================================
FIFA RAG Chatbot — llama3.2:1b
Type 'quit' to exit | 'eval' to run evaluation
============================================================

Question: eval

======================================================================
EVALUATION — Baseline vs SPARQL-generation RAG
======================================================================

[Q1] Which organizations are in the knowledge base?
------------------------------------------------------------
  Baseline (no RAG) ...
  → I'm not sure what a "knowledge base" is, but I can try to provide some general information on organizations that may be part of one. A knowledge base is typically a repositor
y of information, ideas, and expertise gathered from various sources, such a

  SPARQL-generation RAG (LLM) ...

  Gold SPARQL (validated) ...
  → 15 row(s) — http://example.org/fifa/espn_inc | ESPN, Inc.

  LLM RAG : [ERROR] Expected {SelectQuery | ConstructQuery | DescribeQuery | AskQuery}, found 'EXAFI'  (at char 0), (line:1, col:1)
  Repaired: True

[Q2] Which persons are mentioned in the FIFA knowledge base?
------------------------------------------------------------
  Baseline (no RAG) ...
  → I don't have access to a search engine or the ability to look up specific information about individuals. However, I can tell you that the FIFA knowledge base is likely a larg
e database of information related to football (soccer), and it may include e

  SPARQL-generation RAG (LLM) ...

  Gold SPARQL (validated) ...
  → 15 row(s) — http://example.org/fifa/andrew | Andrew

  LLM RAG : [ERROR] Expected SelectQuery, found 'ex'  (at char 25), (line:2, col:3)
  Repaired: True
```

Figure 1 — RAG evaluation: Q1 (organizations) and Q2 (persons) — llama3.2:1b generates invalid SPARQL; Gold SPARQL returns correct results

```
[Q4] What is FIFA linked to in the knowledge base?
------------------------------------------------------------
  Baseline (no RAG) ...
  → In the knowledge base, FIFA (Fédération Internationale de Football Association) is linked to several topics and concepts. Here are a few examples:

1. **Football**: The most obvious connection is that FIFA is closely associated with football, also kn

  SPARQL-generation RAG (LLM) ...

  Gold SPARQL (validated) ...
  → 15 row(s) — http://example.org/fifa/indict | http://example.org/fifa/the_united_states_department_of_justice

  LLM RAG : [ERROR] Expected {SelectQuery | ConstructQuery | DescribeQuery | AskQuery}, found 'EX'  (at char 0), (line:1, col:1)
  Repaired: True

[Q5] Which entities are linked to London?
------------------------------------------------------------
  Baseline (no RAG) ...
  → London is a major global city with numerous interconnected entities. Here are some of the key ones:

1. **Government**: The United Kingdom (UK) Parliament, the House of Commons, and the House of Lords make up the UK government.
2. **Economy**: The fi

  SPARQL-generation RAG (LLM) ...

  Gold SPARQL (validated) ...
  → 11 row(s) — http://example.org/fifa/fifa | http://example.org/fifa/exclude

  LLM RAG : [ERROR] Expected {SelectQuery | ConstructQuery | DescribeQuery | AskQuery}, found 'SKOS'  (at char 0), (line:1, col:1)
  Repaired: True
```

Figure 2 — RAG evaluation: Q4 (FIFA links) and Q5 (London entities) — consistent LLM failure pattern with parasite tokens (EX, SKOS)

```
======================================================================
SUMMARY
======================================================================
Q    LLM correct    Repaired   Question
----------------------------------------------------------------------
Q1   No             True       Which organizations are in the knowledge base?
Q2   No             True       Which persons are mentioned in the FIFA knowledge base?
Q3   No             True       Which places are referenced in the knowledge base?
Q4   No             True       What is FIFA linked to in the knowledge base?
Q5   No             True       Which entities are linked to London?
```

Figure 3 — Evaluation summary: all 5 LLM queries failed, self-repair attempted on all

5.5 Analysis & Failure Cases

The Gold SPARQL column confirms the KB correctly stores organizational, personal, and geographic entities. All 5 reference queries return meaningful results, validating the full KB construction pipeline.

The LLM generation failures follow a consistent pattern: llama3.2:1b systematically generates a spurious token at position 0 (EXAFI, EXTRACT, SKOS, EX) instead of SELECT. This is a structural limitation of 1B parameter models on strict format-following tasks. The self-repair loop corrects syntax but the repaired version also fails for the same underlying reason.

Q4 is the most instructive case: the baseline correctly answers from training data (general FIFA knowledge), while Gold SPARQL reveals KB-specific information (FIFA indictment linked to US Department of Justice) unknown to the LLM. This illustrates the exact scenario where a functional RAG pipeline would add value. A 7B+ parameter model would be required for reliable NL-to-SPARQL generation.

6. Critical Reflection

6.1 KB Quality & Size Impact

The final KB (19,515 triples) falls below the 50k–200k target. With 35 confidently aligned entities as expansion anchors (1,000 triples each), the theoretical ceiling is ~35k triples before deduplication. A local Wikidata ElasticSearch index would provide more anchors and significantly increase expansion volume.

6.2 Predicate Noise

The 1,729 extracted predicates (lemmatized verbs) far exceed the recommended 50–200 relations. Most are generic English verbs (be, have, include) carrying minimal semantic information. Clustering these verbs using word embeddings or filtering to a domain-specific whitelist would produce a cleaner predicate space and directly improve KGE performance.

6.3 Rule-Based vs Embedding Reasoning

SWRL rules produce deterministic, interpretable inferences but cannot generalize beyond their explicit conditions. Embedding models generalize across graph structure but produce probabilistic outputs that degrade on sparse KBs. For this KB size and noise level, rule-based reasoning is more reliable for targeted inferences; embeddings become competitive at scale (>50k triples with clean predicates).

6.4 RAG Limitations

The RAG pipeline infrastructure is correct and functional. The failure is entirely in the LLM generation component: llama3.2:1b lacks the instruction-following capacity to produce syntactically valid SPARQL consistently. This is consistent with known benchmarks — SPARQL generation typically requires 7B+ parameter models for acceptable accuracy on complex schemas.

6.5 What We Would Improve

- Increase crawl coverage to 20–30 pages for richer entity extraction.
- Cluster 1,729 extracted verbs into 50–100 semantic relation types.
- Lower entity linking threshold to 0.6 with manual spot-checking to increase expansion anchors.
- Use a 7B+ parameter model (Llama 3 8B, Mistral 7B) for reliable NL-to-SPARQL generation.
- Add predicate-controlled SPARQL expansion to complement 1-hop expansion.
- Automate predicate alignment using embedding similarity between private and Wikidata property labels.